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NEW-AGE GLOBAL UNIVERSITY FOR LIBERAL EDUCATION

Semester: II

Category: Core

Operating System

Course Code: CS1103

(Theory and Lab)

Course Outline

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Unit – I	04 Hrs
Introduction to Operating Systems: Operating Systems –Definition Types- Functions -Abstract view of OS- System Structures – System Calls- Virtual Machines.	
Unit – II	08 Hrs
Process Management: Process Concepts, Inter-process communication –Threads –Multithreading, Process Scheduling, First-Come, First-Served (FCFS), Shortest Job First, Round Robin (RR), Priority Scheduling.	
Unit – III	06 Hrs
Synchronization and Deadlocks: Synchronization –Semaphores –Monitors, Hardware Synchronization – Deadlocks –Methods for Handling Deadlocks.	
Unit – IV	12 Hrs
Memory Management: Memory Management Strategies –Contiguous and Non-Contiguous allocation –Paging –Virtual memory Management –File System Design and Implementation – Mass Storage Structure –Disk Scheduling and Management, FCFS, SSTF, and SCAN -I/O Systems- Overview of System Protection and Security.	

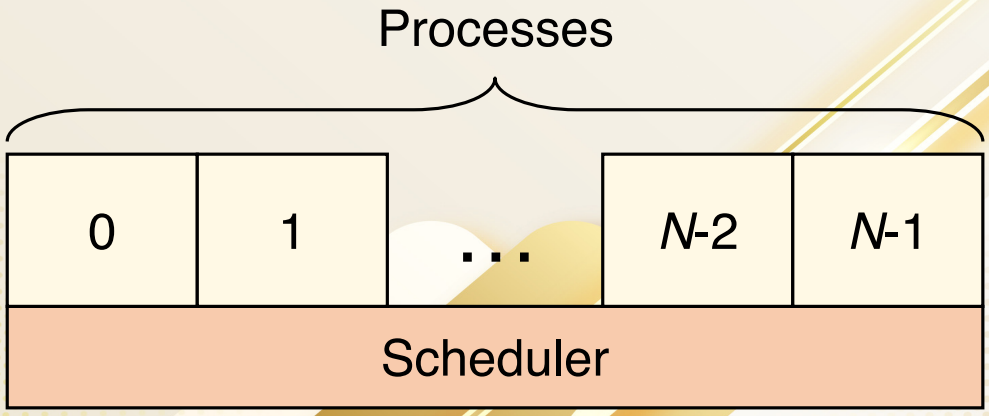
Contents

- Processes in the OS
- Process table entry
- Trap/ interrupt
- Threads: “processes” sharing memory
- Process & thread information
- Thread and stack
- Need for Threads
- Multithreaded Web server
- Interprocess communication
- Socket programming

Created by: Dr. Sangeeta Bhanja Chaudhuri

Processes in the OS

- Two “layers” for processes
  - Lowest layer of process-structured OS handles interrupts, scheduling
  - Above that layer are sequential processes
    - Processes tracked in the *process table*
    - Each process has a process table entry



Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

What is in the process table entry?

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May be stored on stack

Process management
- Registers
- Program counter
- CPU status word
- Stack pointer
- Process state
- Priority / scheduling parameters
- Process ID
- Parent process ID
- Signals
- Process start time
- Total CPU usage

File management
- Root directory
- Working (current) directory
- File descriptors
- User ID
- Group ID

Memory management
- Pointers to text, data, stack
  or
- Pointer to page table

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

Q. What do think, happens in a trap/ interrupt?

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1. Hardware saves program counter (on stack or in a special register)
2. Hardware loads new PC, identifies interrupt
3. Assembly language routine saves registers
4. Assembly language routine sets up stack
5. Assembly language calls C to run service routine
6. Service routine calls scheduler
7. Scheduler selects a process to run next (might be the one interrupted…)
8. Assembly language routine loads PC & registers for the selected process

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

Threads: “processes” sharing memory

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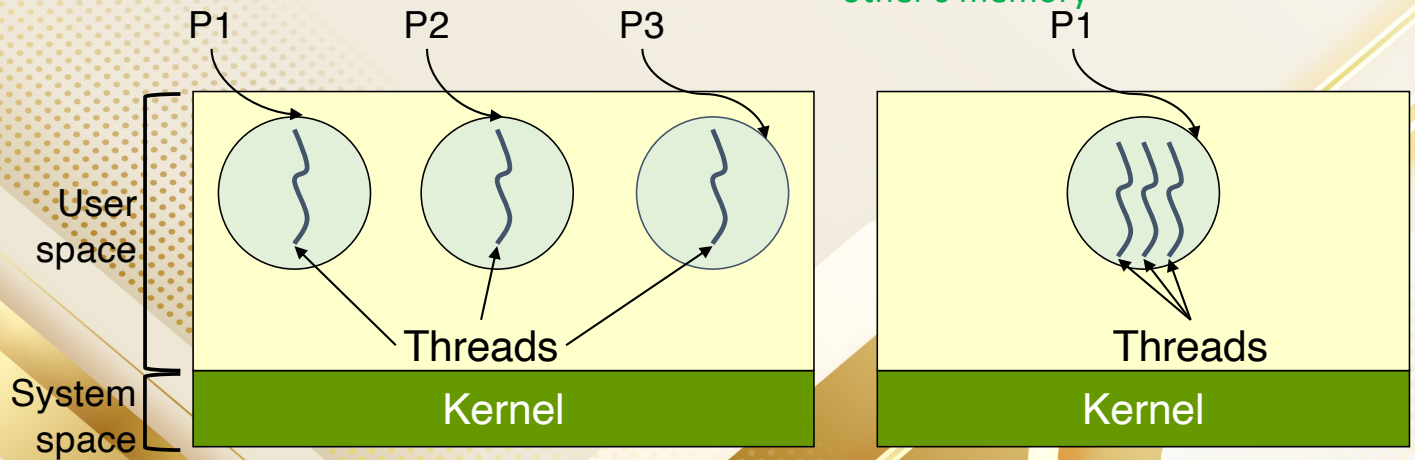
- Process (P) == address space
- Thread == program counter / stream of instructions
- Two examples are shown:
  - Three processes, each with one thread
  - One process with three threads

A thread within a process shares the same address space as the process itself, meaning all threads belonging to a single process can access the same memory locations without needing separate memory allocations for each thread; unlike processes which have their own distinct address spaces and cannot directly access each other's memory



Process & thread information

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Per process items
Address space
Open files
Child processes
Signals & handlers
Accounting info
Global variables

Per thread items	**Per thread items**	**Per thread items**
Program counter	Program counter	Program counter
Registers	Registers	Registers
Stack & stack pointer	Stack & stack pointer	Stack & stack pointer
State	State	State

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

Threads & Stacks

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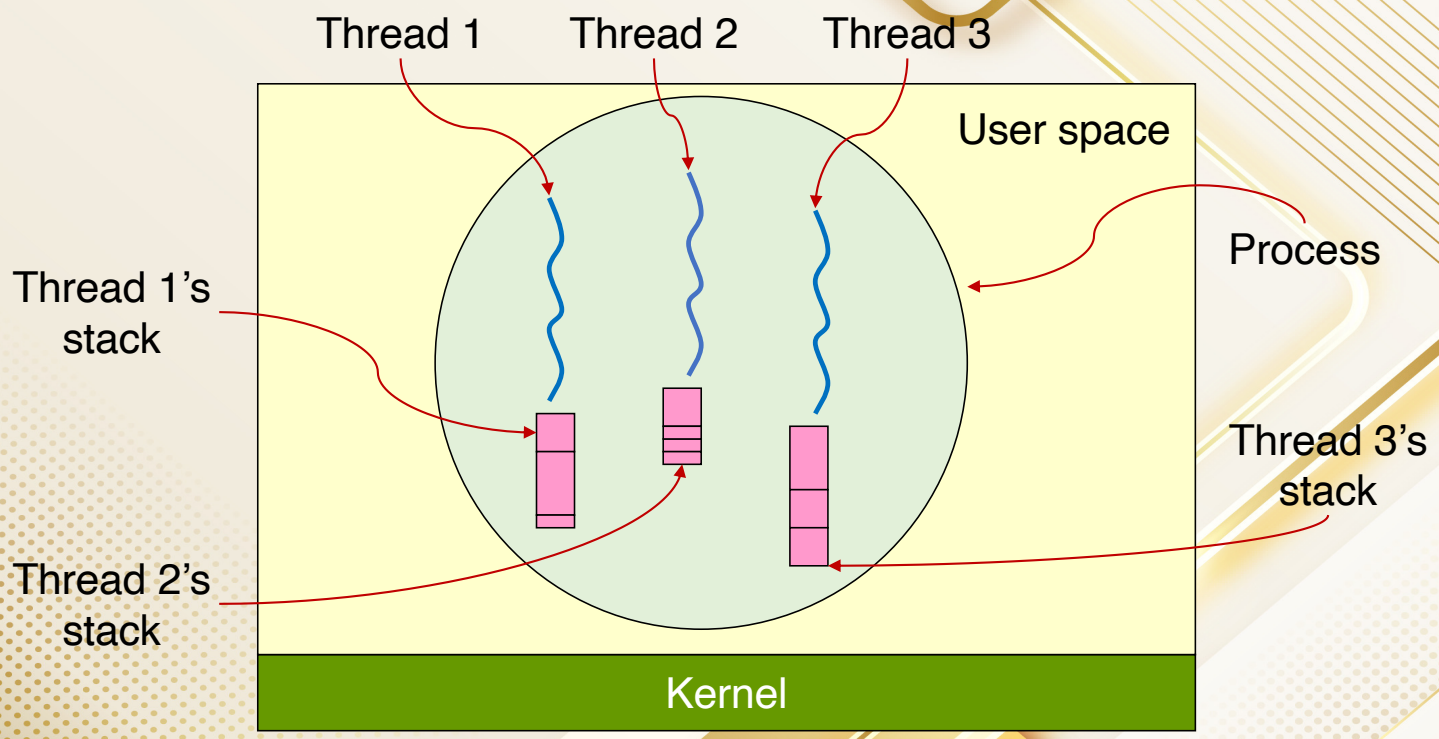

=> Each thread has its own stack!

Q. What is the need for Threads?

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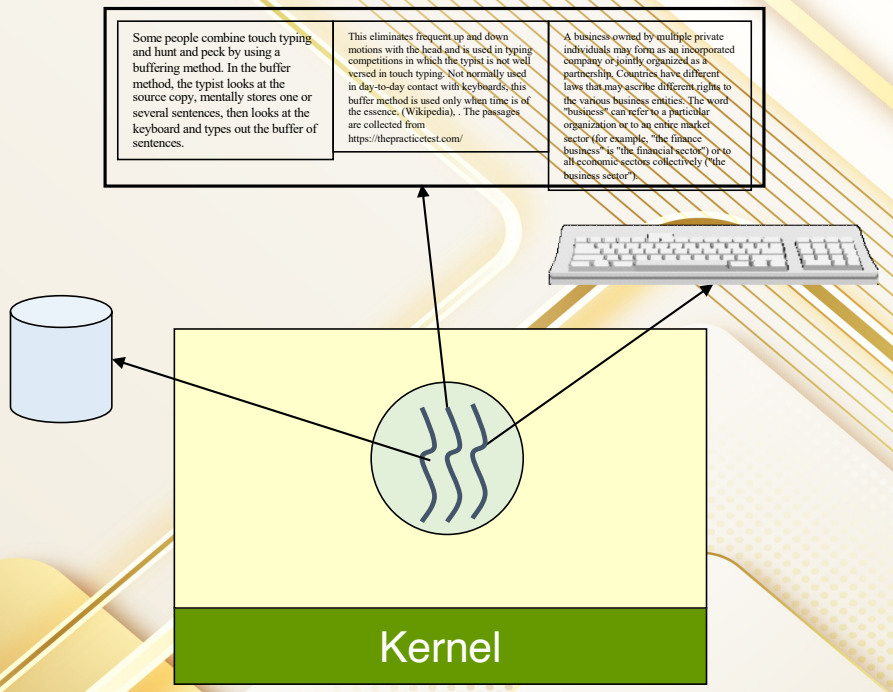
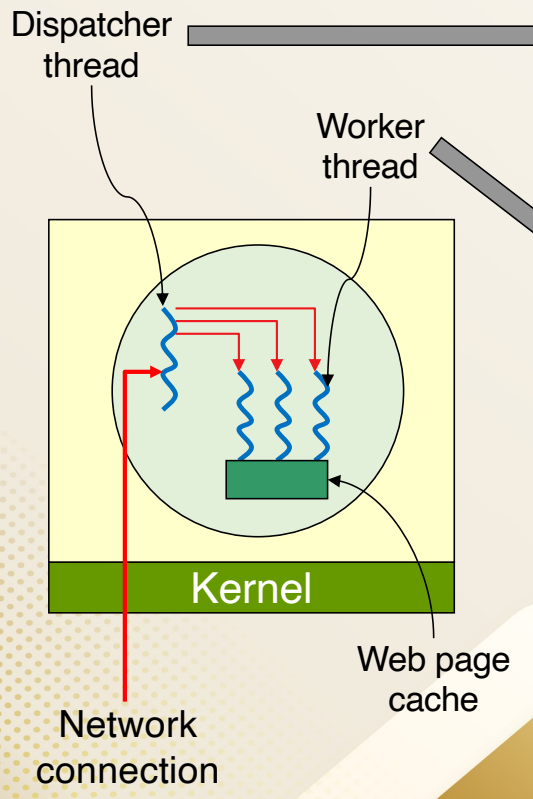
- Achieve parallelism
- Allow a single application to do many things at once
  - Simpler programming model
  - Less waiting
- Threads are faster to create or destroy
  - No separate address space
- Overlap computation and I/O
  - Could be done without threads, but it's harder
- Example: word processor
  - Thread to read from keyboard
  - Thread to format document
  - Thread to write to disk



Multithreaded Web server




```
while(TRUE) {
  getNextRequest(&buf);
  handoffWork(&buf);
}
```

```
while(TRUE) {
  waitForWork(&buf);
  lookForPageInCache(&buf,&page);
  if(pageNotInCache(&page)) {
    readPageFromDisk(&buf,&page);
  }
  returnPage(&page);
}
```

Courtesy: CS 1550, cs.pitt.edu (originaly modified by Ethan L. Miller and Scott A. Brandt)

Explanation

- Infinite Loop (while(TRUE)):This loop runs indefinitely, constantly checking for new requests.
- getNextRequest(&buf);:This function is responsible for fetching the next available request. It could be reading from a queue, network socket, or some other source.
- handoffWork(&buf); :This function delegates the fetched request to another entity (likely a worker thread or a thread pool) for processing.

Explanation in terms of Threading Perspective:

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This code is a typical structure for a producer-consumer model or a thread pool manager:

- Single-Threaded: If this loop runs in a single-threaded environment, all tasks will be handled sequentially, possibly causing delays.
- Multi-Threaded: If handoffWork() hands off the request to a worker thread (e.g., via a thread pool or task queue), this main thread acts as a dispatcher.
- Concurrency: This setup allows multiple worker threads to process requests in parallel, improving efficiency.

Three ways to build a server

- Thread model
  - Parallelism
  - Blocking system calls
- Single-threaded process: slow, but easier to do
  - No parallelism
  - Blocking system calls
- Finite-state machine
  - Each activity has its own state
  - States change when system calls complete or interrupts occur
  - Parallelism
  - Nonblocking system calls
  - Interrupts

Implement Thread

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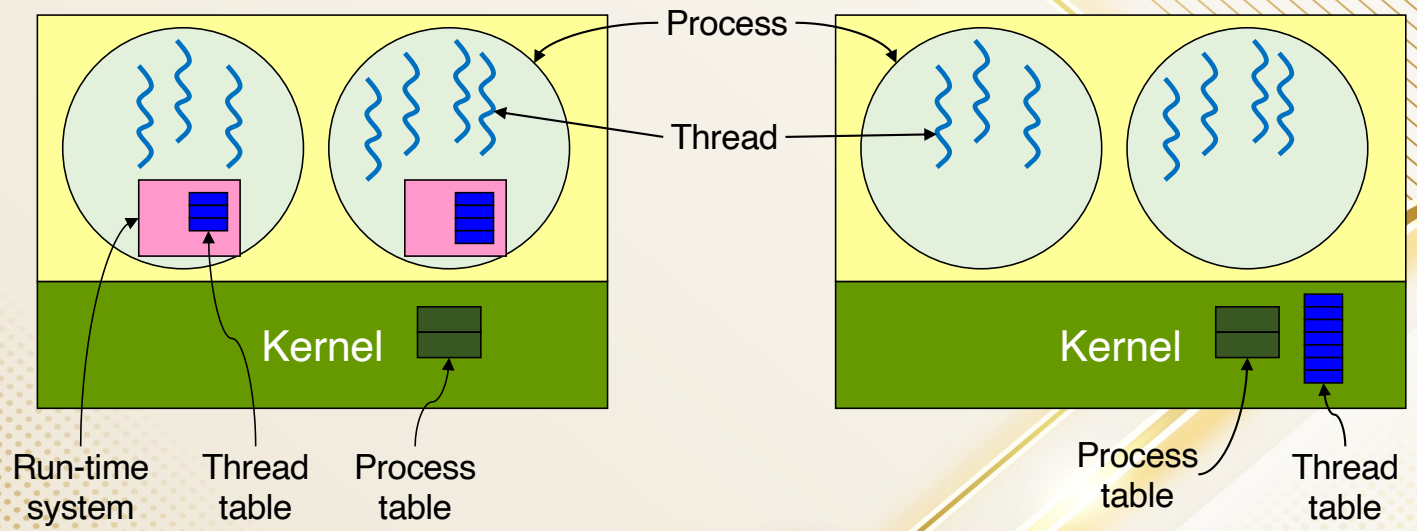

User-level threads
+ No need for kernel support
- May be slower than kernel threads
- Harder to do non-blocking I/O

Kernel-level threads
+ More flexible scheduling
+ Non-blocking I/O
- Not portable

Interprocess Communication

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- Processes within a system may be ***independent*** or ***cooperating***
- Cooperating process can affect or be affected by other processes, including sharing data
- Reasons for cooperating processes:
  - Information sharing
  - Computation speedup
  - Modularity
  - Convenience
- Cooperating processes need **interprocess communication** (**IPC**)
- Two models of IPC
  - **Shared memory**
  - **Message passing**

Activity: Socket programming using TCP

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- Code and documentation Link

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Activity: Socket Programming

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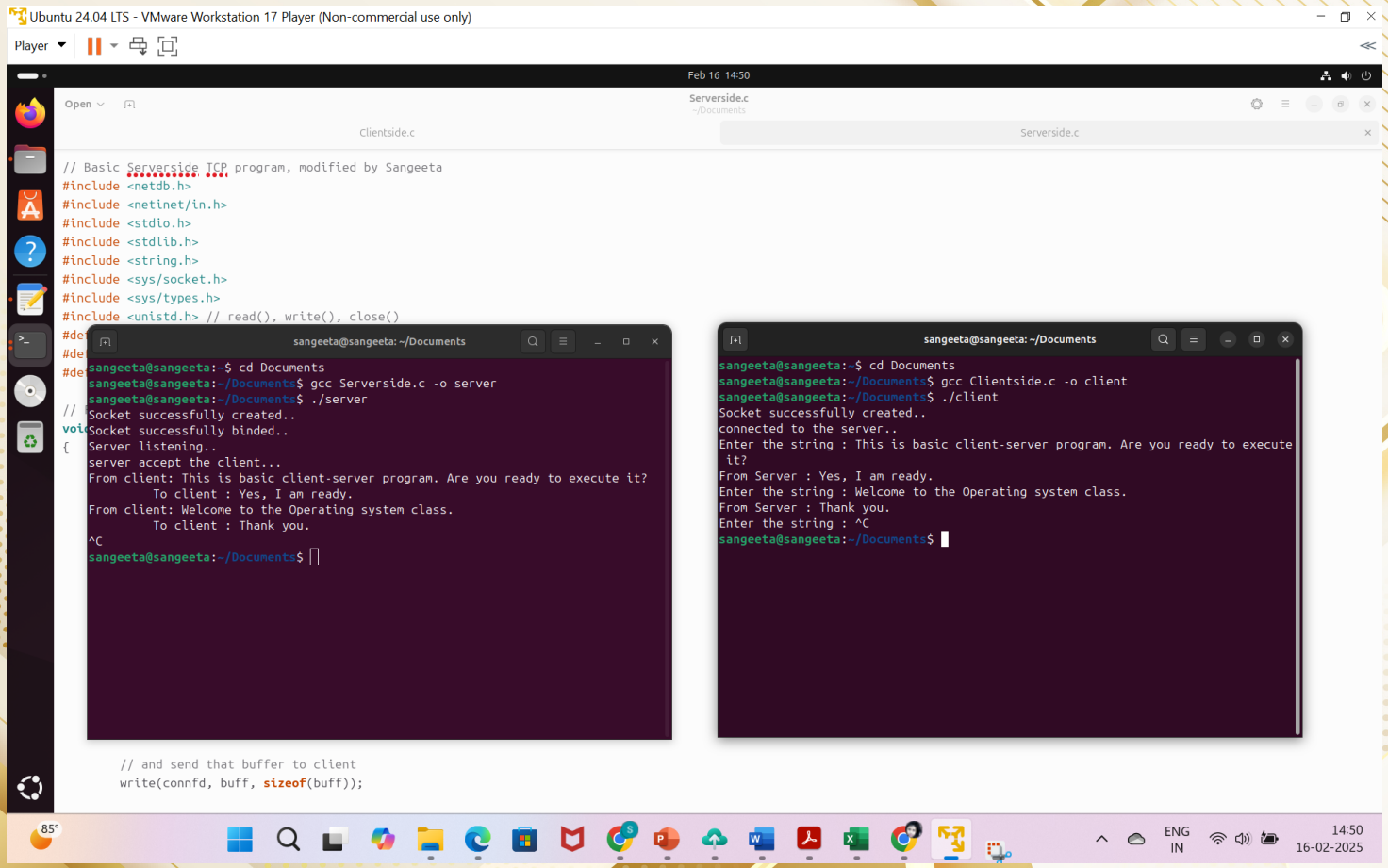

- TCP Server-Client implementation in C

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Server compilation and Execution

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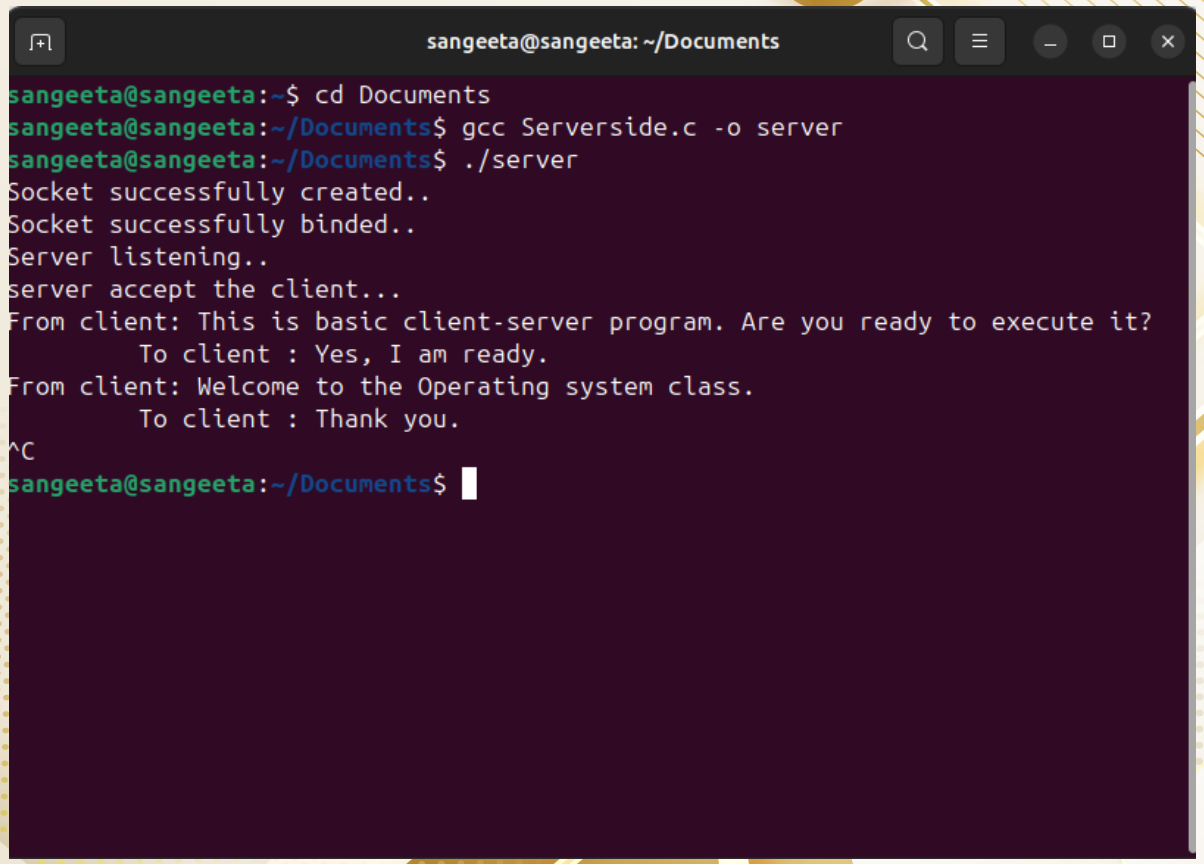

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Client compilation and Execution

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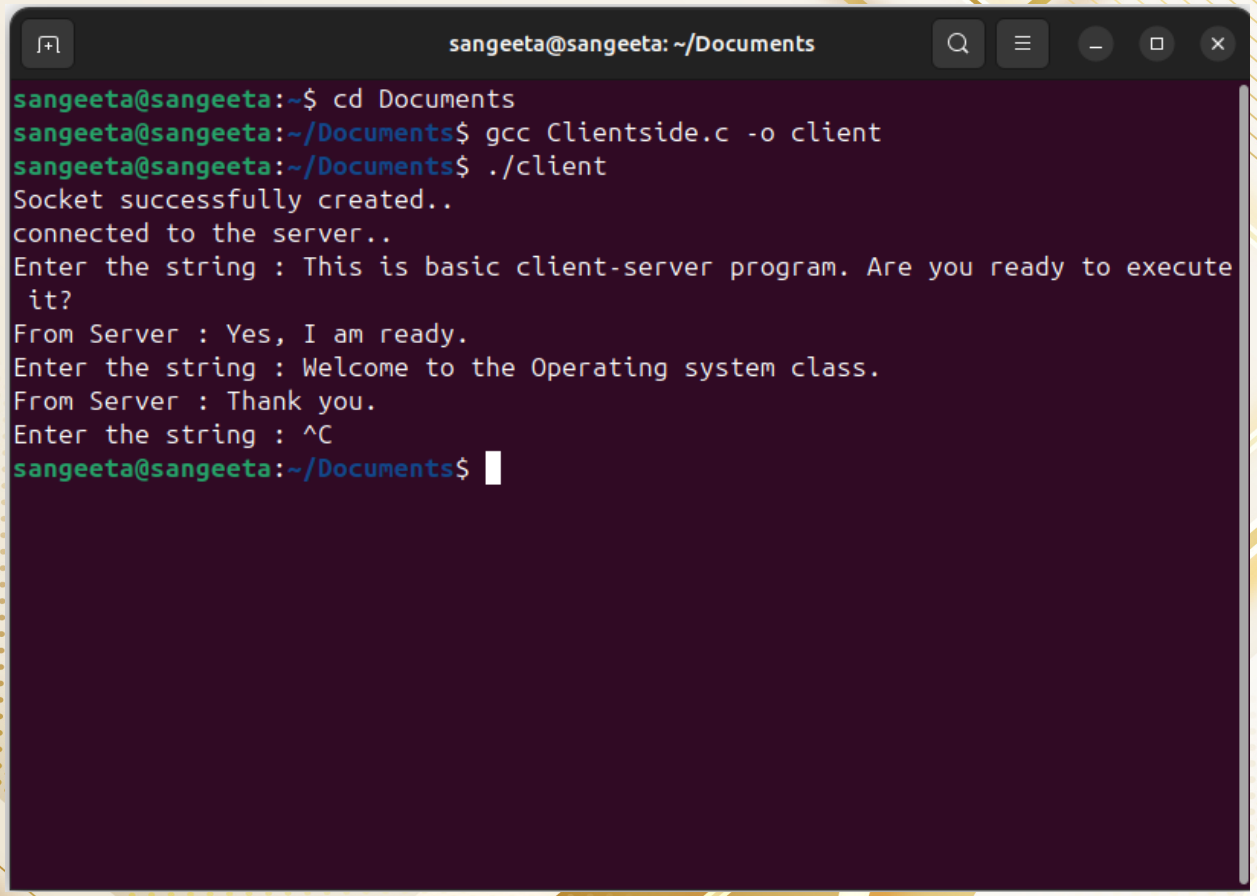

References

- Silberschatz, Galvin, Gagne, Operating System Concepts, John Wiley and Sons, 10th edition, 2023, ISBN: 935746056X
- Modern Operating Systems by Tanenbaum and Bos , 4th edition, Pearson Publisher, ISBN-13: 978-0-13-359162-0
- Courtesy: Prof. Mouli Sankaran, RV University.

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